

MASTER'S COMPREHENSIVE EXAMINATION

Part I Theory (90 minutes) December 6th, 2014

Instructions: Attempt any three out of the following four questions on both pages. Mention which three you would like to have graded. Open book and open notes.

1. a) A barrel at the flea market has 10 sweatshirts. 7 say Meebox and 3 say Sike. If I choose two sweatshirts at random from the barrel, what is the probability that I get one of each?
- b) If 4 Ladidas sweatshirts are added to the barrel and if I choose two, what is the probability that I choose at least one Ladidas?

2. Let X be a random variable whose density is

$$f_X(x) = cx^2(1-x)^2 \quad \text{if } 0 < x < 1 \\ = 0 \quad \text{otherwise}$$

- (a) Find c .
(b) Find $P\{\frac{1}{4} \leq X < \frac{3}{4}\}$.
(c) Find the cdf $F_X(x)$ of X .

3. Let $X_1, X_2, \dots, X_n$ be n independent observations from a normal random variable with mean $-\sigma^2/2$ and variance σ^2 .

Derive the maximum likelihood estimator for σ^2 .

4. It is known that if X_1 and X_2 are iid standard normal random variables then $Y = X_1^2 + X_2^2$ has a chi-square distribution with 2 degrees of freedom. Show that X_1 and X_2 can be written as follows:

$$X_1 = \sqrt{-2 \ln U} \cos(2\pi V) \quad \text{and} \quad X_2 = \sqrt{-2 \ln U} \sin(2\pi V),$$
where U and V are two iid Uniform(0,1) random variables.